

A noninteger subgroup distortion exponent

Conjecture 00000004846

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10 October 2026

The original integer-only polynomial distortion spectrum is false. We give an actual finitely generated group and subgroup with distortion equivalent to the rounded polynomial $n^{3/2}$. This refutes the conjunction without separately resolving its hyperbolic or Baumslag–Solitar assertions.

Let G have eight unrestricted integer coordinates (p, q, r, s, t, a, b, c) . For a second element (P, Q, R, S, T, A, B, C) its product has coordinates

$$(p + P, q + Q, r + R, s + S + pQ, t + T + qR, a + A, b + B, c + C + pT + sR + aB).$$

The identity is zero. The inverse coordinates are

$$(-p, -q, -r, -s + pq, -t + qr, -a, -b, -c + pt + sr - pqr + ab).$$

Lean proves the actual group laws, rather than assuming them. The integral Heisenberg group with product $(a, b, c)(A, B, C) = (a + A, b + B, c + C + aB)$ embeds injectively as the actual subgroup H with $p = q = r = s = t = 0$. The ambient symmetric alphabet consists of the unit elements in the p, q, r, a, b directions and their inverses, denoted X, Y, Z, A, B . The intrinsic alphabet is A, B and their inverses. Actual commutator identities and coordinate normal forms prove generation of every ambient element and every subgroup element.

Use attained minimum lengths in these actual alphabets. The formal definition uses the least natural length of a representing finite word. The complete generation theorems eliminate its outside-generated-range fallback. Finite enumeration of all ambient words of length at most n then gives a finite nonempty subgroup ball and its attained maximum

$$D(n) = \max\{|h|_H : h \in H, |h|_G \leq n\}.$$

Lean proves minimum attainment, exact ball membership, maximum attainment, maximality and monotonicity of this genuine distortion function.

Induction on every ambient word of length L , including negative letters, bounds the five weight-one coordinates by L , s, t by L^2 , and c by $2L^3$. For an embedded subgroup element in radius $n \geq 1$, the correction $d = c - ab$ therefore satisfies $|d| \leq 3n^3$. For a nonnegative central exponent M , use $u = \lfloor \sqrt{M} \rfloor$, $v = M/u$ and $r = M \bmod u$. The actual intrinsic word $[A^u, B^v][A^r, B]$ gives exponent M , including the zero case, with length at most $6u + 6$. Negative exponents use inverse words. Appending the coordinate words gives the all-element uniform bound

$$D(n)^2 \leq 500n^3 \quad (n \geq 1).$$

For the reverse bound, the actual nested commutator $[[X^m, Y^m], Z^m]$ gives the central exponent m^3 in ambient length $10m$. Every intrinsic word of length L has central magnitude at most L^2 . Consequently

$$m^3 \leq D(10m)^2 \quad (m \geq 0), \quad \lfloor \sqrt{\lfloor n/10 \rfloor^3} \rfloor \leq D(n) \quad (n \geq 0).$$

Both inequalities concern genuine minima and attained maxima.

The standard coarse comparison used here is

$$f \preceq g \iff \exists C \geq 1 \forall n, f(n) \leq Cg(Cn + C) + Cn + C.$$

Let $F(n) = \lfloor \sqrt{n^3} \rfloor$. Lean proves the exact rounding characterization $F(n)^2 \leq n^3 < (F(n) + 1)^2$, and both comparisons $D \preceq F$ and $F \preceq D$ with constants 23 and 10 respectively, including every small radius. This is the natural-radius representative of degree 3/2, with full linear rescaling and additive-linear errors.

For degree one a proposed comparison $D \preceq n$ makes the nested commutator family linearly bounded, contradicting its cubic squared lower bound at a sufficiently large m . For every integer $k \geq 2$, a proposed comparison $n^k \preceq D$ implies $n^4 \leq (12000C^5 + 6C^2)n^3$ for every positive n ; choosing $n = 12000C^5 + 6C^2 + 1$ contradicts it. Thus no positive integer power is coarsely equivalent to the actual distortion.

The exported unconditional theorem is

`Coordinate4846.original4846_counterexample`

It combines the faithful actual subgroup, both finite generation witnesses, attained distortion, degree-3/2 coarse equivalence and all positive integer exclusions. Seven source/audit modules were compiled by the author, and the separate complete-type audit reports only `propext`, `Classical.choice` and `Quot.sound`. These author commands do not constitute Root acceptance. Independent original/source/PDF review, Root mechanical verification, official fresh replay and guarded promotion are required separately. No novel-theorem or upstream-acceptance claim is made.